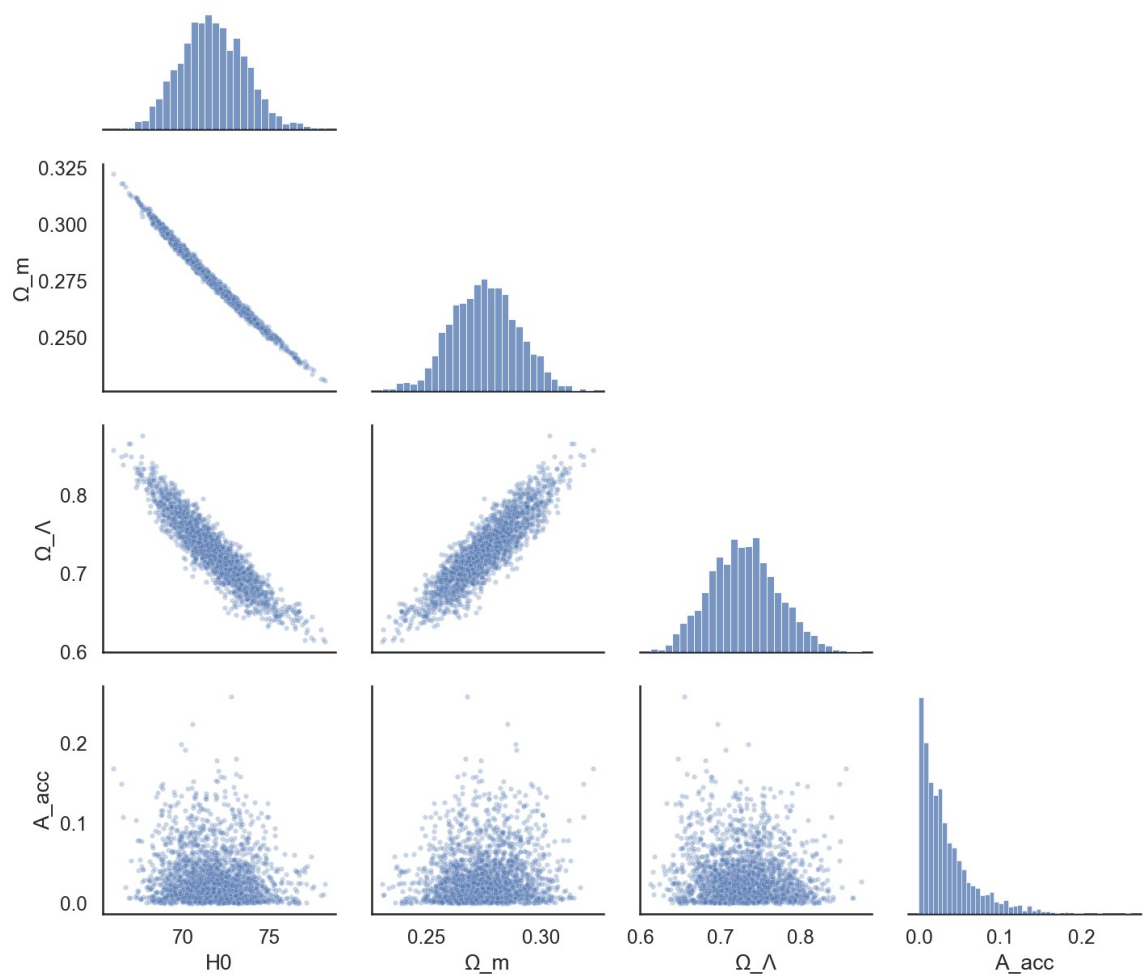


Parameter	Median	-1σ	$+1\sigma$
H_0	71.6382	1.8319	1.9685
Ω_m	0.2753	0.0144	0.0143
Ω_Λ	0.7289	0.0416	0.0443
A_{acc}	0.0243	0.0186	0.0370

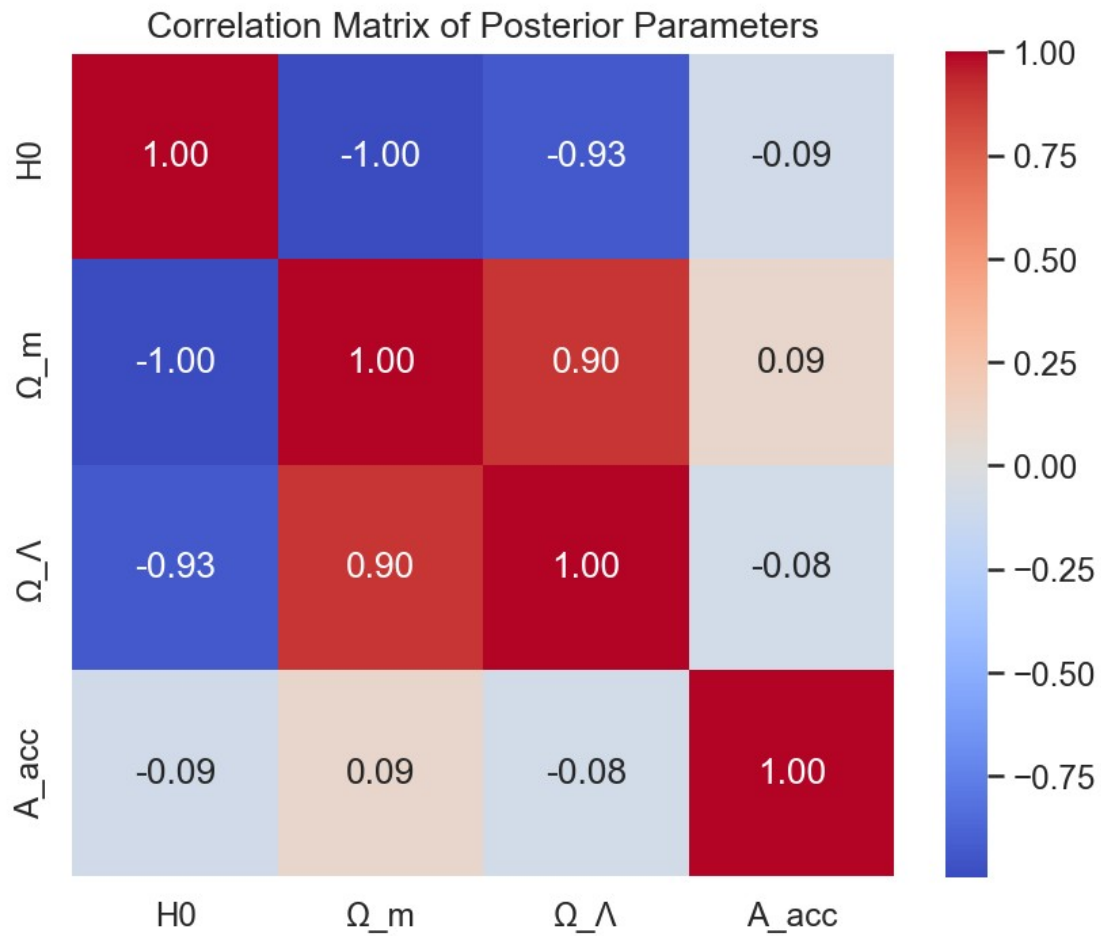
2. Graphical Analysis

Pairwise Posterior Distributions

Pairwise Posterior Distributions



Correlation Matrix



3. Interpretation

- Strong anticorrelation between H_0 and Ω_m , and H_0 and Ω_Λ , indicative of degeneracy in the expansion history.
- Ω_m and Ω_Λ are positively correlated, reflecting the $\Omega_{total} \sim 1$ flatness prior.
- A_{acc} (acceleration term) is weakly correlated with other parameters, suggesting it captures orthogonal acceleration effects.
- A_{acc} 's distribution is positively skewed, indicating a small but non-zero contribution is likely favored.

4. Suggested Next Steps

- Overplot $\mu(z)$ vs. Pantheon+ data using the median posterior parameters.
- Compare this fit to a baseline Λ CDM model (set $A_{acc} = 0$).
- Run PCA or eigenvalue decomposition to analyze parameter space dimensionality.
- Evaluate model residuals and test fit quality using χ^2 or BIC.
- Extend the analysis to include BAO and CMB likelihood terms for tighter constraints.

